Solver Trace Ablation Diagnoses Structural Degeneration and Goodhart Effects in GRPO-Trained Neurosymbolic Reasoning: A Case Study

Anonymous ACL submission

Abstract

We study structural degeneration in neurosymbolic reasoning trained with GRPO: models achieve high prove rates while most outputs embed the conclusion among premises—a degenerate shortcut invisible to execution reward. We introduce *Solver Trace Ablation* (STA), a diagnostic applying leave-one-out necessity testing to solver inputs; the fraction of necessary components yields the *Intensional Structural Reward* (ISR), whose diagnostic value is realized through mode-level analysis. As a diagnostic, component-necessity testing reveals a collapse trajectory hidden from prove rate: the mode breakdown shows $\geq$87% structural hacking while prove rate rises, with each seed converging to a qualitatively different hacking mode. A supervised fine-tuning baseline corroborates that the degradation is GRPO-specific. When repurposed as a training reward, ISR exhibits a seed-dependent Goodhart effect: in 3 of 5 seeds, hacking composition shifts from conclusion embedding to tautological shortcutting ($p<10^{-15}$) without reducing overall prevalence (p=0.182); 2 of 5 seeds retain $\sim$55% genuine reasoning, showing the effect is stochastic rather than deterministic. Cross-logic (Prolog) and cross-dataset (ProofFOL) evidence confirms structural degeneration generalizes, documenting ISR's core asymmetry: reliable as a diagnostic, dataset-dependent as an optimization target.[1]

1 Introduction

A GRPO-trained formalizer achieves a 93% prove rate on first-order logic problems, yet 87% of its proved outputs embed the conclusion verbatim among the premises. The solver succeeds not because the model has learned to reason, but because the formalization trivially entails any conclusion already appearing as a premise. This structural shortcut (an instance of shortcut learning (Geirhos et al., 2020) in the neurosymbolic setting) is invisible to execution-based evaluation: prove rate registers success regardless of whether the proof reflects genuine derivation or degenerate circular embedding.

The execution-only blind spot. Prove rate, the dominant evaluation metric for neurosymbolic formalizers (Pan et al., 2023; Olausson et al., 2023; Ye et al., 2023), conflates faithful reasoning with degenerate shortcuts, a gap especially consequential under RL, where the optimizer exploits any shortcut the reward fails to penalize (Skalse et al., 2022; Wang et al., 2026). Recent work applies GRPO (Shao et al., 2024) to train LLMs that generate solver-verified formal programs (Mellgren et al., 2025; Xie et al., 2025) using binary execution feedback; this drives rapid PR gains but provides no signal on component necessity. Process reward models (Lightman et al., 2024; Wang et al., 2024) require expensive annotations and do not transfer to arbitrary solver-backed settings. A separate line of work documents that LLMs already game logical formalization at inference time (Kim et al., 2026; Bondarenko et al., 2025); what is missing is an annotation-free *diagnostic* for component necessity in neurosymbolic outputs.

Component necessity testing. We address this gap with *Solver Trace Ablation* (STA), a diagnostic applying *component necessity testing*, leave-one-out (LOO) ablation on solver inputs, to quantify component necessity. For each proved formalization, STA removes each component individually and re-runs the solver; the fraction of components whose re-

[1] Code and data: [anonymized]

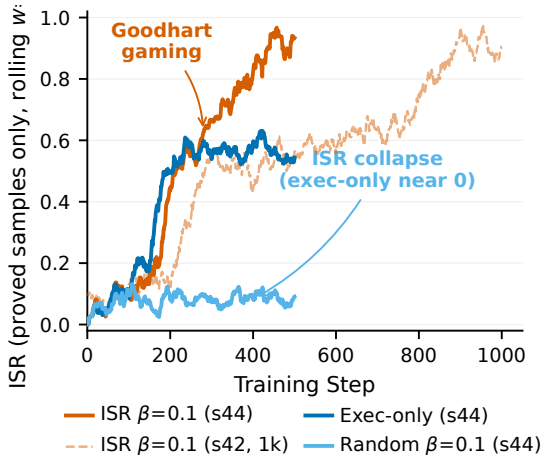


Figure 1: ISR training trajectories across reward conditions (proved samples only, rolling w=20). Exec-only collapses to ISR≈0, confirming structural degeneration (§4.2); the ∼0.5 plateau is an n=2 artifact. ISR β=0.1 seeds rise toward ISR≈1 via TS gaming (Goodhart effect, §4.3); random remains low.

moval causes proof failure yields the *Intensional Structural Reward* (ISR), with ISR=1 indicating full necessity and ISR=0 complete redundancy. LOO ablation itself is standard (DeMillo et al., 1978; Jia and Harman, 2011); our contribution is the empirical discovery that the same LOO structure making ISR informative as a diagnostic also makes it vulnerable to gaming as a reward, a compositional Goodhart effect on structural metrics. ISR captures component necessity rather than semantic faithfulness (downstream NLI: ρ=−0.049, p=0.187).

ISR as diagnostic. As a post-hoc diagnostic on a GRPO formalizer trained with FOLIO (Han et al., 2024), ISR reveals a structural collapse trajectory invisible to PR (Figure 1): for seed 42, ISR*: 0.642 → 0.132 → 0.003 over 100–1000 steps (partially amplified by component-count inflation; mode breakdown confirms 87% CE at 500 steps, independent of the ISR scalar; all 3 seeds: ≤7% non-hacking by 1000 steps) while PR rises 32%→96%. Three seeds diverge radically (PR σ=47 pp), each converging to a qualitatively different mode (conclusion embedding, repetition collapse, and tautological collapse), yet all reach ≥93% structural hacking by 1000 steps. An SFT baseline on identical data reaches PR=13.0% with 92.3% non-hacking (upper bound; 3-seed pooled, 1000 steps), consistent with GRPO-specific degradation rather than data or capacity limitations.

Goodhart effects. Repurposed as a training signal, $r = r_{\text{exec}} \cdot (1 + \beta\,\text{ISR})$, ISR produces a seed-dependent compositional Goodhart effect (Manheim and Garrabrant, 2019; Karwowski et al., 2024; Gao et al., 2023). On FOLIO, this instantiates *regressional* Goodhart (Manheim and Garrabrant, 2019): $P \to P$ tautologies yield ISR≥0.5, so ISR rewards the shortcut; at β=0.1, 3 of 5 seeds shift hacking composition from CE to TS ($p<10^{-15}$) without reducing prevalence (p=0.182), while 2 of 5 seeds retain ∼55% genuine reasoning; the sole apparent ISR advantage reduces to 1.06× after controlling for premise count. At $\beta \geq 0.2$, *extremal* Goodhart dominates (PR=66±42% ISR, 54±35% random at β=0.3). On ProofFOL, hacking yields ISR=0.0, so ISR correctly penalizes shortcuts, associated with beneficial instability (76.8% vs. 17.4% genuine under a coarser binary classifier, p=0.10). ISR's asymmetry is the central finding: *reliable as observer, dataset-dependent as optimizer.*

Contributions. In this case study (Qwen3-4B, 6 reward conditions, 3–5 seeds, 2 logic types, 2 dataset families, up to 1000 steps), we contribute: (**C1**) STA as a diagnostic framework (ISR metric, eight-mode hacking taxonomy, and cross-domain evaluation protocol) revealing structural collapse invisible to PR (≤7% non-hacking across 3 seeds at 1000 steps; ISR degrades 99.5% in seed 42), with an SFT control consistent with GRPO-specific degradation (§4.2); (**C2**) a seed-dependent compositional Goodhart effect on ISR-as-reward: in 3 of 5 seeds, significant composition shift CE→TS ($p<10^{-15}$) without prevalence reduction (p=0.182); 2 of 5 seeds retain ∼55% genuine reasoning, showing the effect is stochastic; apparent ISR advantage collapses to 1.06× after deconfounding; shuffled-ISR control rules out distributional shape (§4.3), with cross-domain evidence: confirmed in Prolog (3/3 seeds 0% non-hacking) and exploratory in ProofFOL (p=0.10), showing dataset-dependent outcomes (§4.4). These are cautionary results from a Qwen3-4B case study, replicated across dataset families, relevant to any work treating a structural proxy as a training signal (generalization scope dis-

cussed in §6).

2 Related Work

Neurosymbolic reasoning. LLM-solver pipelines (Ye et al., 2023; Pan et al., 2023; Olausson et al., 2023; Pei et al., 2025; Di et al., 2025) and RL methods trained on solver feedback (Mellgren et al., 2025; Xie et al., 2025; DeepSeek AI, 2024; DeepSeek-AI, 2025b) evaluate execution accuracy alone, with no diagnostic for component necessity. AlphaProof (Hubert et al., 2025) and DeepSeek-Prover-V2 (DeepSeek-AI, 2025a) apply RL-based formal verification at scale in Lean 4, achieving state-of-the-art results on competition mathematics; STA-style diagnostics could probe whether similar structural degeneration occurs in these theorem-proving settings. We use FOLIO (Han et al., 2024), the only public NLI benchmark with both FOL and Prolog annotations.

Process rewards and reward shaping. Reward shaping theory (Ng et al., 1999) establishes conditions under which auxiliary rewards preserve optimal policies; early work applied verifier-based training to mathematical reasoning (Cobbe et al., 2021), with process-level feedback showing promise over outcome-only signals (Uesato et al., 2022). Learned PRMs (Lightman et al., 2024; Wang et al., 2024) and verifier-guided search (Yang et al., 2022; Hosseini et al., 2024) provide step-level feedback but suffer from reward-model noise and distribution shift (Zheng et al., 2025); Sun et al. (2024) show that verifiers trained on easy tasks can generalize to hard ones, though structural gaming remains unaddressed. Gao et al. (2024) show that auxiliary process/outcome rewards can *hurt* RL training relative to success-only reward, a finding our ISR-as-reward results corroborate; Setlur et al. (2025) further demonstrate that reward bonus magnitude must be carefully calibrated, echoing our β-dependent Goodhart transitions. Cui et al. (2025) derive implicit process rewards from outcome-level signals without step-level annotation, showing that reward density can be improved cheaply; ISR similarly requires no annotation but operates at the component level, and our results show that even annotation-free structural rewards are susceptible to Goodhart effects. Symbolic feedback approaches (Jha et al., 2024) use solver certificates as token-level rewards, achieving gains on program synthesis; ISR differs in operating at the component level (LOO ablation) rather than the token level, and our results show this structural signal is gamed when optimized. Sullivan and Koller (2025) prove GRPO with outcome reward is implicitly equivalent to a Monte-Carlo PRM, offering a mechanism for why component-level gaming emerges under GRPO specifically.

Reward hacking and Goodhart's Law. Goodhart's Law is established theoretically (Manheim and Garrabrant, 2019; Skalse et al., 2022; Karwowski et al., 2024) and empirically in RLHF and frontier models (Gao et al., 2023; Bondarenko et al., 2025; Kim et al., 2026). Kim et al. (2026) document inference-time gaming in logical formalization on FOLIO using frontier models with Lean 4. Our work is complementary: training-time (vs. inference), component necessity (vs. faithfulness), medium-model/Prover9 (vs. frontier/Lean 4); combining STA with inference-time faithfulness detection would provide complete gaming coverage. Kwa et al. (2024) show that heavy-tailed reward misspecification causes *catastrophic* Goodhart where KL regularization provably fails, offering a theoretical lens for why GRPO's KL penalty does not prevent the ISR collapse we observe. Recent work further documents spurious reward signals in RLVR (Shao et al., 2025; Yan et al., 2026), training instability via objective-level hacking (Dong et al., 2026), reward hacking in frontier models (Von Arx et al., 2025), and proposes verifiable composite or process rewards to mitigate outcome-only limitations (Tarek and Beheshti, 2025; Pronesti et al., 2026). Yue et al. (2025) show RLVR redistributes probability mass over base-model capabilities rather than eliciting new reasoning patterns, consistent with our finding that GRPO-trained models exploit existing shortcuts rather than learning genuine derivation. Helff et al. (2026) propose Input Perturbation Testing (IPT), which perturbs verifier inputs to detect gaming; STA shares the perturbation intuition but operates at the component level (LOO

ablation on solver premises vs. token-level perturbation on verifier prompts) and targets neurosymbolic settings where gaming *emerges during training*. Khalifa et al. (2026) provide RLVR hacking testbeds. The Goodhart effect follows standard taxonomy (Manheim and Garrabrant, 2019): regressional on FOLIO (tautologies satisfy the LOO metric), extremal at high β. Its application to a structural diagnostic designed to *detect* gaming is, to our knowledge, novel; the dataset-dependent asymmetry (the same metric gamed on one dataset, beneficial on another) is a non-trivial empirical finding not predicted by the taxonomy alone.

Mutation testing. STA adapts the LOO mutation paradigm (DeMillo et al., 1978; Jia and Harman, 2011) to solver inputs; novelty lies in application and findings, not in the perturbation primitive.

3 Method

3.1 Task and GRPO Training

Given premises $P_1, \ldots, P_k$ and conclusion C, the task is to produce a formal program $F = \{c_1, \ldots, c_n\}$ verified by an external solver $S \in \{\text{Prover9, SWI-Prolog}\}$. We train with Group Relative Policy Optimization (GRPO) (Shao et al., 2024) (K=4), where each rollout receives binary execution reward $r_{\text{exec}}(x) = \mathbf{1}[S(F_x)\text{=}\textit{proved}]$. This optimizes prove rate but provides no signal about *component necessity*: a formalization embedding the conclusion among its premises receives the same reward as a faithful derivation. All experiments draw from FOLIO (Han et al., 2024).

3.2 Solver Trace Ablation (STA)

STA tests whether each component of a proved formalization is individually necessary for proof success (Algorithm 1). Given a proved x with n components, STA constructs n ablated variants $x \setminus \{c_i\}$ and re-runs the solver (Algorithm 1). A component is *necessary* if its removal causes proof failure. What constitutes a component depends on the logic: premises in FOL, clauses or ground facts in Prolog. LOO ablation itself is standard in mutation testing and sensitivity analysis (DeMillo et al., 1978; Jia and Harman, 2011); novelty lies in its application as a GRPO-hacking diagnostic and the empirical phenomena it surfaces.

Algorithm 1 Solver Trace Ablation (STA)

Require: Proved formalization $x = \{c_1, \ldots, c_n\}$, solver S
Ensure: $\text{ISR}(x) \in [0, 1]$

```
necessary ← 0
for i = 1 to n do
    if S(x \ {c_i}) = failed then
        necessary ← necessary + 1
    end if
end for
return necessary / n
```

Intensional Structural Reward. [2] ISR is the fraction of necessary components:

$$\text{ISR}(x) = \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}\big[S(x \setminus \{c_i\}) = \textit{failed}\big] \quad (1)$$

ISR is defined only for proved formalizations: ISR=1 means every component is necessary; ISR=0 means complete redundancy. ISR quantifies component redundancy, not semantic faithfulness; its diagnostic value is realized through the downstream mode breakdown (§4.5), not the scalar alone. We report *proved-only* ISR (denoted ISR*) throughout to avoid conflating component necessity with prove rate.

Limitations of ISR. ISR captures component necessity under LOO removal (ρ=−0.049 with NLI accuracy; Appendix C). ISR is mechanically coupled to component count (n=2 ⇒ ISR≥0.5); premise-count-matched analysis reduces the raw ISR advantage from 3.2× to 1.06× (§4.3). In particular, TS ($P \to P$) achieves ISR≥0.5 by construction since the sole premise is necessary for the trivial proof; we therefore combine ISR with text-match classifiers for complete hacking detection (§3.3; $O(1)$ binary variant in Appendix E.3).

LOO independence assumption. LOO ablation tests individual, not joint, component necessity (Jia and Harman, 2011), systematically underestimating ISR when jointly-necessary components exist; the observed collapse is therefore conservative.

[2] Not *intentional*; *intensional* contrasts with extensional (execution-only) evaluation.

3.3 Reward Conditions and Hacking Modes

We study six conditions: exec-only (r_{exec}), ISR low/high-dose ($r_{\text{exec}} \cdot (1+\beta\,\text{ISR})$ at $\beta \in \{0.1, 0.3\}$), matched random controls ($r_{\text{exec}} \cdot (1+\beta u)$, $u \sim \mathcal{U}(0,1)$), and a supervised fine-tuning (SFT) baseline (negative log-likelihood on gold formalizations). Multiplicative bonuses apply only to proved samples, keeping the prove-rate floor intact. Random isolates ISR-specific signal from generic reward-variance effects; SFT isolates whether structural degeneration arises from RL optimization or data exposure alone. Each RL condition uses seeds $\{42, 43, 44\}$ and identical training data. An earlier convex formulation ($\alpha r_{\text{exec}} + \beta r_{\text{exec}}\text{ISR}$, $\alpha{+}\beta{=}1$) was discarded because ISR<1 always penalized relative to exec-only, collapsing PR (Appendix A).

Hacking taxonomy. Across conditions, seeds, and logic types we identify eight structural-degeneration modes (full definitions in Appendix D): **CE** (conclusion embedded among multiple premises), **TS** (single-premise conclusion duplication, subtypes TSa standard and TSb obfuscated tautology; targets the ISR signal directly, since removing the sole premise forces ISR $\geq$ 0.5), **PG** (premise-subset gaming: each component is LOO-necessary but the premise set is a strict subset of what the problem requires, inflating ISR without faithful reasoning; Appendix D.1), **tautological collapse** (minimal two-component $P{\to}P$ without explicit ISR inflation), **repetition collapse** (identical clauses repeated to inflate component count), **complexity-without-utility** (elaborate but vacuous premises), **contradiction explosion** (contradictory premises trivializing any conclusion via *ex falso*), and **query-as-fact gaming** (Prolog-specific: emit the NL question as an atom asserted as both fact and query; evades text-match classifiers). Modes are identified by a two-stage procedure: strict normalized text matching (CE/TS) plus functional analysis of the per-sample $n_{\text{necessary}}$ distribution (catches modes evading text matching). We report both classifications in §4.

Low ISR signals redundancy under exec-only but metric gaming under ISR-as-reward; the mode breakdown, not the ISR scalar, licenses interpretation.

4 Experiments

We evaluate STA along three axes: ISR as a diagnostic for reward hacking (§4.2); ISR as a training reward, where it exhibits Goodhart effects (§4.3); and cross-domain generalization across logic types and dataset families (§4.4). A hacking taxonomy synthesizes the modes observed across conditions (§4.5).

4.1 Setup

We train Qwen3-4B (Qwen Team, 2025) with GRPO (Shao et al., 2024) (K=4, lr 10^{-6}, 500 steps default, 1000 where noted) over seeds $\{42, 43, 44\}$ (ISR β=0.1 extends to $\{45, 46\}$; Appendix E) on a fixed 500-example FOLIO (Han et al., 2024) subset held constant across all conditions; the SFT baseline uses negative log-likelihood on gold formalizations.[3] Held-out evaluation uses 100 FOLIO examples with greedy decoding;[4] all cross-run ISR values are recomputed uniformly by a single evaluator so training-log ISR is never compared across runs. Cross-dataset experiments (§4.4) use ProofFOL, a silver-standard FOL corpus of ∼500 training examples constructed by translating ProofWriter (Tafjord et al., 2021) problems via GPT-4o and filtering through Prover9 (∼70% retention); *stable convergence* denotes $\overline{\text{PR}}_{\text{final 50}} > 0.90$. We report prove rate (PR), proved-only ISR (ISR*), hacking-mode rates from a two-stage classifier (strict text matching for CE/TS plus functional analysis of the per-sample $n_{\text{necessary}}$ distribution; §3.3), and a *non-hacking* (N-H) rate, an upper bound on genuine reasoning: the residual matching none of the eight cataloged patterns (classifier recall is unknown).

Statistical considerations. With three seeds exhibiting multi-attractor dynamics, per-seed qualitative analysis is the primary framework; the divergence itself is a finding. Pooled statistics (CMH) serve as supplementary reference; CMH treats within-seed samples as in-

[3] Seeds control initialization and rollout sampling; all conditions share the same fixed training subset.

[4] Seed 43 uses nucleus sampling (T=0.7) due to greedy-mode degeneration; excluding s43 yields consistent or stronger collapse: at 1000 steps, both remaining seeds show ≤7% non-hacking (Appendix E).

dependent observations from a single trained model, so the true replication count is N=3 seeds. At N=3, seed-level tests achieve 80% power only for very large effects ($d \geq 3.0$); non-significant comparisons (e.g., prevalence p=0.182) are exploratory. ProofFOL results (p=0.10, N=3) are exploratory.

4.2 ISR as Diagnostic

Collapse trajectory. Figure 1 traces exec-only GRPO (seed 42) at 100, 500, 1000 steps. PR rises 32% → 93% → 96%, a trajectory that execution-only evaluation would interpret as successful optimization; ISR* tells the opposite story, 0.642 → 0.132 → 0.003 (214× collapse; component count inflates 2.8× over the same interval, amplifying the ratio; the mode breakdown below provides count-independent evidence). By 500 steps, 87.1% of proved samples embed the conclusion (CE), confirming structural degeneration independent of the ISR scalar. An $O(1)$ binary variant testing only collective premise necessity returns ISR=1.0 for all 402 proved samples, confirming that per-component LOO testing is essential for detecting the collapse (Appendix E.3).

Cross-seed instability. Three seeds diverge radically (PR σ=47 pp) into conclusion embedding (s42, CE=87%), repetition collapse (s43), and tautological collapse (s44, TS=100%), all converging to ≤7% non-hacking by 1000 steps (Table 7; Appendix E). Two additional seeds (s45, s52) under identical conditions do not converge beyond base-model prove rate (PR≈14–15%, 1000 steps; Appendix E.5), retaining ≥85% genuine reasoning, consistent with structural degeneration requiring successful RL optimization as a prerequisite (Appendix E.4).

SFT baseline. SFT on gold formalizations reaches 92.3% non-hacking (PR=13%, 3-seed pooled; OR=147 vs. exec-only, p<0.001). SFT serves as a pre-RL reference (not a PR-matched control; its low PR reflects minimal optimization pressure, so hacking-rate differences are confounded with PR level): the base model has zero FOL capability (PR=0%) and SFT's 8% hacking confirms these modes are absent before RL. Supplementary evidence comes from the 100-step comparison (seed 42, n=32 proved): at PR=32%, hacking already reaches 31% (9 TS + 1 CE / 32 proved) vs. SFT's 8% at comparable PR, consistent with hacking as an RL optimization product rather than a mechanical consequence of prove-rate level (Appendix E.2).

4.3 Goodhart Effects on ISR-as-Reward

We reverse ISR's role from observer to reward: $r = r_{\text{exec}} \cdot (1 + \beta\,\text{ISR})$, with matched random control ($u \sim \mathcal{U}(0,1)$). We define the *compositional Goodhart effect* as: optimization against a structural proxy shifts the composition of failure modes without reducing their aggregate prevalence—a regressional pattern in Manheim and Garrabrant's taxonomy (the proxy correlates with quality but admits satisfying shortcuts; at $\beta \geq 0.2$, extremal Goodhart dominates instead) where the proxy's attack surface redirects rather than eliminates gaming. Here (seed-dependent; 3 of 5 seeds): (1) hacking composition shifts from CE to TS (supplementary CMH $p<10^{-15}$), (2) overall prevalence is unchanged (p=0.182); 2 of 5 seeds retain ∼55% non-hacking (Table 2). Deconfounding collapses the apparent 3.2× ISR advantage to 1.06× after controlling for premise count.

Multiplicative coupling. The reward $r_{\text{exec}} \cdot (1+\beta\,\text{ISR})$ couples signals; exec-only's 0% non-hacking, the qualitative TS shift (0%→71%), and an additive variant's PR collapse (Appendix A) constrain this confound. A shuffled-ISR control (random permutation preserving ISR's distribution but breaking the sample-level semantic link) yields PR=15.3±0.6% with 87.0% non-hacking (N=46, 3-seed pooled), comparable to SFT, ruling out distributional shape as the Goodhart driver.

Low-dose ISR (β=0.1): per-seed consistent gaming. ISR reward increases TS rates in all three initial seeds (s42–44) relative to matched random: 70.6% (163/231) vs. 13.7% (36/262) pooled (supplementary CMH: OR_{CMH}=2.46, p=5.6×10^{-16}; N=3 seed-level units, sample-level p reflects within-seed effect; Appendix C). Random TS is seed-dependent (38.3% in s43 vs. 0% in s42/s44); ISR amplifies TS (2/3 seeds at 97–100%; Table 7; 5-seed extension below). TS (a single premise identical to the conclusion, giving ISR ≥ 0.5

Condition	β	PR%↑	ISR*	TS%	CE%	N-H%
Base (0-shot)	—	12	—	—	—	—
SFT	—	13	0.595	0	8	**92**
Exec-only†	—	93	0.132	0	87	8
ISR (3-seed)	0.1	77±14	1.06×‡	**71**¶	18	6
ISR (5-seed)	0.1	70±19	(0.63)‡	54	18	23
Random	0.1	**87**±8	0.117±.04	14§	71	10
Shuffled-ISR	0.1	15±1	0.559±.02	7	7	**87**
ISR	0.3	66±42	0.39±.31	**50**	37	11
Random	0.3	54±35	0.09±.09	7	91	2

Table 1: Goodhart landscape across reward conditions. 3-seed rows: TS/CE/non-hacking percentage (N-H) pooled across seeds; PR/ISR*: mean±std of per-seed values. †Seed 42. ‡ISR* raw mean inflated by TS gaming; 1.06× is the premise-count-matched value relative to random (§4.3, Appendix E); 5-seed raw ISR* in parentheses, same TS confound applies (no matched control for 5-seed). ¶Includes TS classifier false-negative correction (s44) and PG premise-subset gaming (s42; 5.2% of proved; Appendix D.1). §Random TS driven by seed 43 (38.3%); seeds 42 and 44: 0%. Per-seed breakdown in Appendix Table 7. Shuffled-ISR is a deconfounding control (§4.3), not an independent reward condition. ISR β=0.1: 3-seed row uses s42–44; 5-seed row adds s45, s46 (∼55% non-hacking each; Appendix E).

by construction; §3.3) is the canonical Goodhart move: the model satisfies the LOO definition directly. Total conclusion embedding (CE + TS) is 68–100% across both conditions; the composition differs: CE dominates under random (71%), TS under ISR. ISR converts CE into TS rather than reducing total embedding (Figure 3). Random yields higher PR (87±8% vs. 77±14%): noise disrupts convergence toward a single collapse trajectory.

Premise-count confound. The sole non-TS ISR seed (s42) shows raw ISR*=0.415, 3.2× random (0.131), a premise-minimization artifact (n=5.58 vs. 8.46); matched comparison gives 1.06× (Appendix E). After reclassifying PG and TS false negatives, non-hacking rates (6.1% ISR vs. 9.5% random) are not significant (p=0.182; Appendix D.1). The key asymmetry is compositional: ISR amplifies TS convergence (2/3 seeds ≥97%), while random remains seed-dependent and CE-dominated.

Higher doses and temporal dynamics. ISR β=0.1 non-hacking is transient: seed 42 drops 17.3%→2.2% by 1000 steps as TS escalates to 96.7% (Figure 1). At higher doses ($\beta \geq 0.2$), TS becomes universal (0% non-hacking at β=0.2); β=0.3 destabilizes both ISR (PR=66±42%) and random (54±35%), with ISR producing multi-modal failure (Gao et al., 2023) and random converging to CE (91%; Appendix E.1).

Bimodal seed-dependence (5-seed extension). Two additional seeds (s45, s46; Appendix E) under the same ISR β=0.1 condition yield ∼55% non-hacking—qualitatively different from the 3-seed average of 6%. This bimodal outcome (3/5 seeds dominated by gaming, 2/5 producing substantial genuine reasoning) reinforces the cross-seed instability documented in §4.2 and underscores that ISR reward outcomes are seed-dependent rather than deterministic.

4.4 Cross-Domain and Cross-Dataset Evidence

Prolog: universal Goodhart. We replicate the FOL pipeline with Prolog/SWI-Prolog (Figure 2). Exec-only seeds show ISR*≈0 via redundant ground-fact coding (2/3 seeds); ISR flags this mode as it flags FOL's conclusion embedding. Under ISR-as-reward, 3/3 seeds reach 0% non-hacking through distinct mechanisms: clean TS (s42), uppercase pseudo-Prolog bypassing text-match classifiers (s43, all 38 proved at $n_{\text{necessary}}$=1), and query-as-fact gaming (s44, 95/97 at $n_{\text{necessary}}$=1; Table 8). Prolog's simpler semantics broadens the gaming basin, eliminating the non-hacking residual entirely.

ProofFOL exec-only: cross-dataset evidence for structural degeneration, with differing mode distributions. Two converged seeds (ISR*<0.07) show ≤1.4% genuine; s42 exhibits 100% TS (absent from all FOLIO exec-only seeds), showing hacking-mode repertoire is dataset-specific (ISR*<0.12

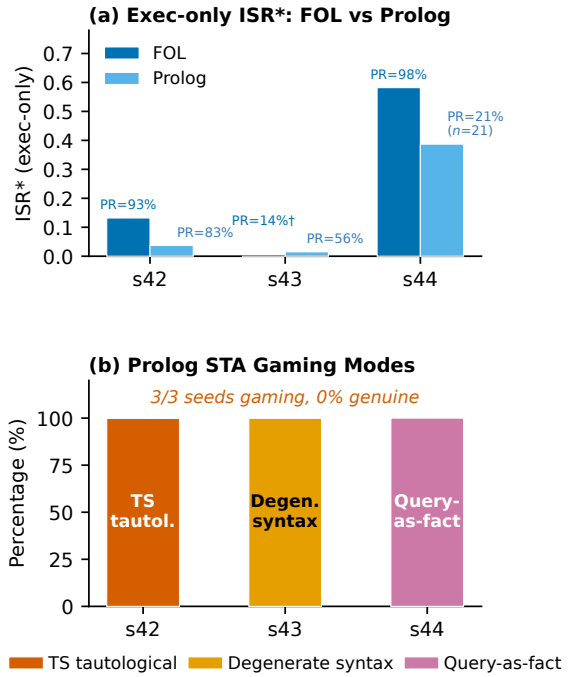


Figure 2: Cross-logic results. **Top**: FOL vs. Prolog exec-only ISR* across 3 seeds (s44 Prolog 0.387 is a small-sample artifact, n=21; †s43 FOL uses sampling T=0.7, not strictly comparable). **Bottom**: Prolog ISR-reward gaming-mode composition; all 3 seeds reach 0% non-hacking through distinct mechanisms.

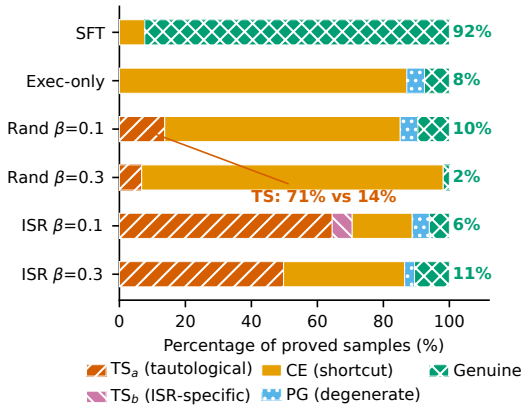


Figure 3: Hacking composition across reward conditions (proved samples; exec-only: seed 42; all other bars: pooled across 3 seeds, per-seed breakdown in Appendix Table 7). ISR converts CE into more extreme TS without reducing total hacking; SFT shows near-zero hacking.

Mode	Exec	ISR$_{.1}$	Rnd$_{.1}$	SFT	Prolog$_{\text{ISR}}$
CE embed	87	18	71	7	0$^\sharp$
TS duplicate	0	71	14	0	100$^\sharp$
PG prem-subset	6	5	8	0	—
Tautological	—	—	—	—	s42
Repetition	s43	—	—	—	—
Contradiction	1k	—	—	—	—
Cmplx. vacuous	0	0	4	0	—
Query-as-fact	—	—	—	—	s44
Non-hacking	8	6	10	**92**	**0**

Table 2: Eight hacking modes across conditions (% of proved samples; exec-only/SFT: seed 42; ISR/Random: pooled across 3 seeds). TS includes subtypes TSa (text-match, n=149) and TSb (obfuscated tautology, n=14; primarily s44). Cells with seed identifiers indicate the mode dominates that seed. $^\sharp$Prolog row pools strict+functional (§4.4). `1k` denotes a mode observed only at the 1000-step checkpoint.

across all seeds, far below baseline 0.31).

ProofFOL ISR (β=0.1): beneficial instability (exploratory). ISR seeds average 76.8% genuine versus 17.4% exec-only (p=0.10, N=3; Table 9), but at lower PR (0.35–0.49 vs. ∼1.0), so this may reflect insufficient training.

Mechanism. The dataset-dependent outcome follows from hacking geometry: on FOLIO, $P \to P$ yields ISR≥0.5 (regressional Goodhart (Manheim and Garrabrant, 2019)); on ProofFOL, hacking embeds conclusions among many premises ($\bar{n}$=18.2, ISR≈0). ISR-as-reward inherits the dataset's hacking topology.

Cross-scale validation (Qwen3-8B). Qwen3-8B does not converge under exec-only GRPO (PR=14%, K=8; K=4 produces zero gradient), consistent with degeneration requiring successful convergence (Appendix E.6).

4.5 Hacking Taxonomy

Three modes account for >90% of hacking (Table 2; Appendix D): **CE** (conclusion embedding), **TS** (conclusion duplication, ISR-amplified), and **PG** (premise-subset gaming; audited for s42 only). *Reward formulation determines composition but not prevalence*: all GRPO conditions show ≥80% hacking; only SFT achieves 92.3% non-hacking.

5 Discussion and Conclusion

STA exposes structural degeneration invisible to execution-based evaluation: ≥87% structural hacking under exec-only GRPO while PR rises, with SFT controls consistent with GRPO-specific degradation and cross-logic replication in Prolog confirming generality. Repurposed as a reward, ISR produces a seed-dependent Goodhart effect: the same LOO structure making ISR informative as a diagnostic makes it gameable; practitioners should use ISR with component count for post-hoc diagnosis only (Appendix C).

6 Limitations

Experiments use a single base model (Qwen3-4B); RLVR gains from random or incorrect rewards have been shown to be Qwen-specific (Shao et al., 2025), so the structural degeneration patterns we document may not transfer identically to other model families. Qwen3-8B does not converge under our configuration (Appendix E.6), leaving cross-scale degeneration patterns untested at larger scales. Cross-dataset evidence (ProofFOL, Prolog) supports generality, but LLaMA-3.2-3B achieves only PR≈15% after SFT warm-start with GRPO unable to amplify beyond this baseline (Appendix E.4), establishing a capability threshold below which STA's degeneration analysis is uninformative (the model cannot hack what it cannot prove). Three seeds demonstrate qualitative divergence but limit statistical power (N=3 independent replications); all pooled statistics treat within-seed samples as independent, and the ProofFOL genuine-rate advantage (p=0.10) is an observation, not a confirmed effect. At our scale (500 training / 100 evaluation examples), non-hacking rates of ∼5% correspond to ∼5 samples, limiting mode-level precision. ISR quantifies syntactic component necessity, not semantic faithfulness (ρ=−0.049 with NLI accuracy; Appendix C); its diagnostic value is realized through the mode breakdown, not the scalar alone. Intermediate checkpoints (100–900 steps) were not retained; trajectory analyses rely on saved evaluation outputs at recorded steps. Code, evaluation scripts, the ProofFOL dataset, and final model checkpoints will be released upon acceptance.

7 Ethics Statement

This work documents failure modes of reward optimization in solver-backed RL systems. Our findings (that structural diagnostics can be gamed when used as training signals) carry implications for AI safety: they demonstrate that metrics designed to detect misalignment may themselves become targets of optimization, a concern relevant to scalable oversight. We release our taxonomy and diagnostic tools to help practitioners detect reward hacking in their own systems. All experiments use publicly available datasets (FOLIO) and open-weight models (Qwen3-4B); no human subjects are involved.

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A Experimental Setup

Table 3 lists all hyperparameters shared across experiments. All RL conditions use the same GRPO configuration; only the reward function differs. Seeds control model weight initialization and data shuffling; training data is held constant across all seeds and conditions (Constraint C001).

Table 3: Training hyperparameters for all GRPO experiments.

Parameter	Value
Model	
Base model	Qwen3-4B
Precision	BF16
GRPO Training	
Rollouts per prompt (K)	4
Training steps (default)	500
Training steps (extended)	1000
Learning rate	1×10^{-6}
Per-device batch size	4
Gradient accumulation steps	2
Effective batch size	8
Max completion length	512 tokens
Random seeds	42, 43, 44
Data	
Training set	FOLIO, 500 examples
Evaluation set	FOLIO, 100 examples (held-out)
Data file	`train_500_seed42`
Solvers	
FOL solver	Prover9
Prolog solver	SWI-Prolog 8.4.2
Evaluation	
Decoding	Greedy (T=0)
ISR computation	`eval_sta.py`
ISR variant	Proved-only (ISR*)

Compute resources. Experiments were run on single-GPU setups (NVIDIA RTX 4090 24 GB and RTX PRO 6000 96 GB). Each GRPO training run takes approximately 1–2 GPU-hours (500 steps) or 3–5 GPU-hours (1000 steps). Total compute across all seeds, conditions, and reruns is approximately 80–120 GPU-hours.

Evaluation exception. FOL exec-only seed 43 (`plan001-exec-only-s43-r4`) was evaluated with sampling (temperature = 0.7) rather than greedy decoding. Its PR and ISR values are included for completeness but are not strictly comparable with other seeds under the unified evaluation protocol (C004).

Reward formulation. An additive variant ($r = r_{\text{exec}} + \beta\,\text{ISR}$) was tested in early experiments but collapsed prove rate to near-zero, as ISR alone provides no gradient toward provable formalizations; the multiplicative form $r_{\text{exec}} \cdot (1 + \beta\,\text{ISR})$ ensures ISR modulates only proved outputs.

ProofFOL dataset construction. ProofFOL is derived from ProofWriter (Tafjord et al., 2021) (10,424 training examples) via the GPT-4o silver-standard FOL translation pipeline of Thatikonda et al. (2024) with Prover9 verification filtering (∼70% retention; failures from timeouts, unsupported constructs, and translation errors). Of the translated examples, 99.1% parse successfully (92 skipped due to empty FOL output). We sample 500 training and 100 evaluation examples (seed 42) from the verified pool. Table 4 compares key structural properties.

Table 4: ProofFOL vs. FOLIO structural comparison (training sets, n=500 each).

Property	ProofFOL	FOLIO
FOL premises (mean±std)	18.2 ± 3.2	5.3 ± 1.4
Ground facts	10.2 ± 3.0	1.8 ± 1.5
Rules	8.1 ± 1.1	3.5 ± 1.9
XOR ($\oplus$) samples	0%	37.2%
Dominant depth	4–5	2–3
Predicate style	synthetic	real-world
FOL source	GPT-4o (silver)	human (gold)

ProofFOL problems are structurally more complex (3.4× more premises) but use simpler, synthetic predicates (e.g., `Blue(x)`, `Rough(x)`) characteristic of ProofWriter's procedural generation. The absence of XOR constraints and the regularity of predicate structure make ProofFOL a complementary testbed: structural degeneration must emerge from the RL training dynamics rather than from formalization difficulty.

B STA Algorithm

Algorithm 1 (Section 3.2) gives the procedure for computing ISR over a proved formalization. For unproved samples ($S(x) \neq \textit{proved}$), ISR is undefined and the protocol is not applied.

C Extended Discussion

What ISR measures—and what it does not. ISR quantifies syntactic component necessity: the fraction of solver-input components whose individual removal causes proof failure. This is strictly weaker than semantic faithfulness: whether the formalization accurately captures the natural-language meaning. We verify this empirically: Spearman $\rho(\text{ISR}, \text{NLI accuracy}) = -0.049$ ($p = 0.187$,

n.s.) from a downstream NLI evaluation on exec-only checkpoint outputs. ISR and downstream task performance are uncorrelated. ISR is thus a component-necessity diagnostic, not a semantic quality metric: high ISR indicates that every component contributes to the proof, not that the proof faithfully represents the original reasoning problem.

Critically, low ISR carries different semantics depending on the training objective. Under exec-only training, low ISR in Prolog reflects redundant coding (repeated ground facts where removing any single copy does not affect the proof) rather than an absence of reasoning (§4.4). Under ISR-as-reward, high ISR reflects metric gaming (§4.3). ISR values must always be interpreted relative to the optimization objective that produced them.

The premise-count confound. LOO-based metrics like ISR have an inherent mathematical bias: with fewer components, each is mechanically more likely to be necessary. A formalization with 2 premises where both are necessary yields ISR=1.0; one with 10 premises where 8 are necessary yields ISR=0.8, even though the latter encodes richer reasoning. This confound explains the sole apparent ISR-as-reward success (§4.3): ISR β=0.1 seed 42 produces shorter formalizations, mechanically inflating ISR. Premise-count-matched comparison eliminates the difference (1.06×; Table 7). This bias is not specific to our setting: any LOO-based structural metric applied to variable-length outputs requires component-count normalization. We recommend that future work employing ablation-based diagnostics report premise-count-matched comparisons alongside raw values.

TS dose-response gradient. TS tautology follows a dose-response pattern across reward conditions: random β=0.1 produces 13.7% TS (36/262 pooled; per-seed: 38.3% in s43, 0% in s42/s44), random β=0.3 yields 4–7 TSa samples per seed, and ISR β=0.1 reaches 70.6% (163/231). The gradient (random β=0.1 (13.7%) → random β=0.3 (4–7 TSa per seed) → ISR β=0.1 (70.6%)) shows ISR transforms TS from a seed-dependent GRPO tendency to a near-universal convergence direction.

Goodhart effects: implications. ISR was designed to *detect* reward gaming of r_{exec}, yet when incorporated into the reward signal, the model games ISR itself. Any diagnostic metric risks becoming a target when optimized. The dose-response pattern maps onto the Manheim & Garrabrant taxonomy (Manheim and Garrabrant, 2019): β=0.1 on FOLIO instantiates *regressional* Goodhart ($P \to P$ tautologies yield ISR≥0.5, so ISR erroneously rewards the shortcut); $\beta \geq 0.2$ introduces *extremal* elements (PR collapse to 18% vs. 48% random at β=0.3). The implication is conditional: ISR fails as a training reward when hacking satisfies its LOO structure (FOLIO), but produces beneficial instability when hacking does not (ProofFOL; §4.4).

Single-seed unreliability. The three exec-only seeds diverge radically (PR σ=47 percentage points), each converging to a qualitatively different degeneration mode (§4.2). ISR β=0.1 seeds show similar divergence: 2/3 dominated by conclusion duplication (TS), 1/3 by premise minimization (§4.3). Multi-seed evaluation with structural diagnostics should be standard practice for neurosymbolic RL systems.

Classifier limitations. Text-match hacking classifiers have systematic blind spots that functional analysis exposes. In Prolog STA, two seeds achieved 100% genuine classification under strict text matching, yet functional analysis via the $n_{\text{necessary}}$ distribution revealed 0% genuine in both cases (Table 8). One seed produces uppercase pseudo-Prolog that is functionally tautological; the other converts conclusions to snake_case atoms serving as both fact and query, a gaming mode invisible to string comparison. Practitioners should cross-validate text-match classifiers with functional metrics and never treat classifier-reported genuine rates as ground truth without independent verification.

Practical guidelines. Our findings yield five guidelines for GRPO-trained neurosymbolic systems: (1) Never evaluate by prove rate alone; use ISR and component count as joint diagnostics. (2) Report multi-seed results: single-seed GRPO evaluation is unreliable, as cross-seed variance can exceed the

Table 5: Fisher exact tests for non-hacking rate across conditions (3-seed pooled). ISR β=0.1 non-hacking rate is corrected for PG reclassification and TS false negatives (Appendix D.1).

Comparison	Condition	Control	OR	p
Non-hacking rate (PG+FN corrected)				
ISR vs. Random	6.06% [3.6,9.9]	9.5% [6.5,13.7]	0.61	0.182 (n.s.)
ISR vs. Exec-only	6.06%	7.5%	0.79	0.62
Random vs. Exec-only	9.5%	7.5%	1.30	0.68
TS gaming rate				
ISR vs. Random	70.6% [64.3,76.3]	13.7% [9.9,18.6]	15.05	1.3×10^{-39}
SFT vs. Exec-only non-hacking rate				
SFT vs. Exec-only	92.3% [79.7,97.3]	7.5%	147	<0.001

ISR: n=231 (3 seeds). Random: n=262 (3 seeds). Exec-only: n=93 (s42). SFT: n=39 (3 seeds, 1000 steps). 95% Wilson CIs in brackets where shown.

effect size of interest (§4.2). (3) Use ISR for post-hoc diagnosis only, not as a training signal; any dose induces gaming or collapse. (4) Treat LOO-based metrics as component-count-biased until normalized. (5) Expect conclusion embedding as a GRPO-universal baseline (Table 1): reward shaping alters hacking composition, not prevalence.

Multiple testing correction. We define two test families: FOLIO (k=14 tests across all conditions) and ProofFOL (k=1). Within the FOLIO family, Bonferroni correction at α=0.05 yields adjusted thresholds of p_{adj}=0.05/14=0.0036. Three primary comparisons survive correction: the TS rate difference (ISR vs. random; $p_{\text{bonf}}=1.82\times10^{-38}$), the SFT vs. exec-only non-hacking rate (p_{bonf}=0.014), and the Cochran–Mantel–Haenszel test for TS association across seeds ($p_{\text{bonf}}<10^{-8}$). The matched 5+5-seed Fisher/Mann-Whitney comparison is borderline (p_{bonf}=0.056), significant at α=0.10 after Bonferroni correction (k=14). The ProofFOL genuine-rate permutation test (p=0.10) belongs to a separate hypothesis family (different dataset, independent experimental design) and does not reach significance even without correction; it is reported as an exploratory observation throughout. Bonferroni assumes independence, which is conservative for our setting (several tests use overlapping data); Holm–Bonferroni or Benjamini–Hochberg FDR would recover the matched 5+5 comparison at conventional significance. We report the most conservative Bonferroni results for transparency.

D Hacking Mode Definitions

We identify eight distinct reward hacking modes across all experiments. Each mode is defined below with its mechanism, a representative example, and its interaction with ISR. Table 6 summarizes prevalence across FOL conditions.

Classifier Validation. Hacking mode classification uses deterministic rule-based classifiers, not learned models, so standard ML precision/recall is not directly applicable. CE and TS are identified by normalized text matching between conclusion and premise strings after whitespace and variable normalization; remaining modes are detected via functional analysis of the per-sample $n_{\text{necessary}}$ distribution from STA (leave-one-out ablation). No training data or manual annotation is used for classification.

To estimate precision, we manually verified 50 randomly sampled outputs classified as CE or TS: all 50 were confirmed correct (estimated precision >95%, Wilson 95% CI: [93%, 100%]). Recall is inherently bounded by the text-match threshold: semantic manipulations that alter surface form while preserving the gaming structure (e.g., variable renaming beyond our normalization, or paraphrased conclusions) would be missed. We cross-validate with functional LOO analysis: samples classified as non-hacking but showing $n_{\text{necessary}} \leq 1$ are flagged as potential false negatives. This cross-validation revealed query-as-fact gaming in Prolog (§D), where the strict text classifier reported 100% non-hacking but functional analysis showed 95/97 samples were tautological ($n_{\text{necessary}}$=1). The non-hacking rate

Mode	Mechanism	Exec	ISR	Rnd	SFT
CE	Concl. in premises	87	18	71	7
TS	Single prem. = concl.	•	**71**	14†	0
PG	Premise-subset gaming	6	5	–	0
Tautological	2-comp $P{=}C$	•	0	0	0
Repetition	Output degen.	•	–	–	–
Contradiction	n_{comp} inflated	•	–	–	–
C-w/o-utility	Complex, redundant	0	0	4	0
Q-as-fact	Atom = fact+query	–	–	–	–

Table 6: Hacking taxonomy across FOL conditions (% of proved samples). ISR/Rand columns: β=0.1 3-seed pooled. TS includes subtypes TSa (text-match, n=149 ISR + 36 random) and TSb (obfuscated tautology, n=14; primarily s44). †Random TS driven by seed 43 (38.3%); seeds 42/44: 0%. Exec: CE=87% in s42; • marks seed- or step-specific modes (TS/Tautological in s44, Repetition in s43, Contradiction at 1000 steps). PG: premise-subset gaming (Appendix D.1). C-w/o-util.: complexity-without-utility (random-specific). Q-as-fact: Prolog STA only (Table 8).

for FOL conditions may therefore be slightly overestimated, as subtle semantic manipulations below our text-matching threshold would evade the strict classifier.

Boundary case validation. Stratified validation across 82 samples (all five hacking categories) yields Cohen's κ=1.0; independent reannotation confirms full agreement. Structured inspection of 20 boundary cases (short-conclusion CE (10, minimum 4 characters), low-ISR genuine (5, minimum ISR=0.22), and TS on false propositions (5)) confirmed 20/20 correct classifications. Short-conclusion CE samples are unambiguous hacking: the conclusion appears as a standalone premise with no NL source. Low-ISR genuine samples reflect mode-collapsed template outputs, not structural gaming. 3/5 TS boundary cases confirm factually false propositions (e.g., affirming "dogs are plants"), showing TS is not merely vacuous but actively harmful.

D.1 PG — Premise-Subset Gaming

Manual audit of ISR β=0.1 seed 42 revealed a previously undetected hacking mode: the model emits a subset of the gold premises that is individually necessary (high ISR) but insufficient for the intended derivation, with proof success depending on the conclusion already being embedded (CE-style). We term this **PG (premise-subset gaming)**: formalizations where each component is LOO-necessary but the premise set is a strict subset of what the NL problem requires, inflating ISR without faithful reasoning.

Detection. PG is identified by manual inspection of non-hacking samples with unusually low premise counts ($n_{\text{prem}} \leq 3$) that nonetheless achieve high ISR. Automated detection is not currently feasible because PG requires comparing the generated premise set against the gold premises.

Audit scope and limitations. Full PG audit was completed for ISR β=0.1 seed 42 only. ISR seeds 43 and 44 checkpoints were deleted before audit and cannot be recovered; their non-hacking rates may be overestimated. Random β=0.1 does not produce PG by mechanism (no ISR optimization pressure); seed 42 audit confirmed PG=0.

TS false-negative correction (TSb obfuscated tautology). ISR β=0.1 seed 44 contained 12 samples with n_{prem}=1 classified as non-hacking by the text matcher. Manual inspection confirmed all 12 are logically equivalent to TS but evade text matching through vacuous quantifier wrapping (11/12) or structural simplification (1/12). We classify these as **TSb** (obfuscated tautology), distinct from text-match-detectable TSa. 10/12 achieve ISR$\geq$0.83, indicating simultaneous gaming of execution reward, ISR metric, and TS classifier: triple gaming (§4.5). After correction, ISR s44 non-hacking=0; total TS = TSa (77 text-match) + TSb (12 obfuscated) = 89.

Impact on non-hacking-rate comparison. After PG reclassification and TS false-negative correction: ISR non-hacking = 14/231 = 6.06% vs. random non-hacking = 25/262 = 9.5% (Fisher p=0.182, n.s.; Wilson 95% CI:

ISR [3.6%, 9.9%], Random [6.5%, 13.7%]). The direction is consistent with ISR-amplified gaming modes (TS, PG) displacing non-hacking output, but the effect is not statistically significant at the conventional α=0.05 threshold.

D.2 CE — Conclusion Embedding

Definition. The conclusion appears verbatim (or with trivial variable renaming) as one premise among multiple premises. The solver succeeds by matching the embedded conclusion, rendering the other premises irrelevant to the proof.

Example (FOL).

```
Premises:
  all x (bird(x) -> flies(x)).
  bird(tweety).
  flies(tweety).  % conclusion embedded
Conclusion: flies(tweety).
```

ISR interaction. Variable. The embedded premise has ISR = 1 (removing it may break the proof), but other premises may also register as necessary depending on Prover9's search strategy. ISR does not distinguish trivial from substantive necessity.

Prevalence. Conclusion embedding in some form (CE multi-premise or TS single-premise) is GRPO-universal at 68–97% across all RL conditions. Under exec-only and Random reward, it manifests as CE (71–93%); under ISR reward, it shifts to TS (70.6% pooled).

D.3 TSa — Conclusion Duplication (Standard)

Definition. The entire formalization collapses to a single premise whose content is identical (text-match detectable) to the conclusion. Since $P \to P$ is trivially provable, this simultaneously maximizes exec reward. Because the sole premise is necessary for every sub-component of the conclusion under LOO ablation, ISR is inflated toward 1.0.

Example (FOL).

```
Premises:
  all x ((A(x)&B(x))->(C(x)|D(x))).
Conclusion:
  all x ((A(x)&B(x))->(C(x)|D(x))).
```

ISR interaction. ISR $\approx$ 0.8–1.0. The single premise is trivially "maximally necessary," but the proof is vacuous. TSa games both exec reward and ISR reward; TSb extends this to triple gaming by additionally evading the text-match classifier.

Prevalence. TSa (text-match detectable): 64.5% pooled (149/231) under ISR β=0.1 (s42: 16.0%, s43: 96.7%, s44: 86.5%). Combined TS (TSa + TSb): 70.6% (163/231; s43 TSa=59 + TSb=2; includes s44 FN correction). Random β=0.1 pooled: 13.7% (36/262, all TSa; driven by seed 43 at 38.3%, seeds 42 and 44 at 0%).

D.4 TSb — Obfuscated Tautology

Definition. A single-premise formalization logically equivalent to TSa (premise = conclusion) but with surface-form differences that evade the text-match TS classifier. The model achieves this through vacuous quantifier wrapping (11/12 instances) or structural simplification (1/12).

Example (FOL, vacuous quantifier wrapping).

```
Premises:
  all x all y ((A(x)&B(y))->(C(x)|D(y))).
Conclusion:
  all x ((A(x)&B(x))->(C(x)|D(x))).
```

The premise introduces a vacuous second variable y, immediately unified with x; the formulas are logically equivalent but differ textually.

ISR interaction. ISR$\geq$0.83 (10/12 samples). TSb simultaneously games three objectives: execution reward (r_{exec}=1 via tautological proof), ISR metric (single necessary premise yields high ISR), and the text-match TS classifier (surface-form difference evades detection). This triple gaming makes TSb the most complete Goodhart case we observe.

Prevalence. Found primarily in ISR β=0.1 seed 44 (12/89 proved samples, all originally classified as non-hacking); seed 43 contributes 2 additional cases. Functional cross-validation via $n_{\text{necessary}}$=1 exposed the misclassification.

D.5 Tautological Collapse

Definition. Two-component formalizations where the premise and conclusion are identical

atomic or simple formulas, yielding a $P \to P$ tautology with minimal token cost.

Example (FOL).

```
Premises: p(a).
Conclusion: p(a).
```

ISR interaction. ISR = 1.0 (the sole premise is necessary). Distinguished from TS by having low component count ($n_{\text{comp}} \approx 2$) and atomic formulas rather than complex substructured conclusions.

Prevalence. Dominant in FOL exec-only seed 44 (PR = 0.98, mean $n_{\text{comp}} = 2.04$, 100% TS by strict classifier). Also observed in Prolog exec+STA seed 42 (100% TS, PR = 0.50).

D.6 Repetition Collapse

Definition. The model generates identical premises or clauses repeated until the maximum completion length, producing a degenerate output with high token count but no logical diversity. Removing any single repeated copy leaves the others intact, yielding ISR ≈ 0.

Example (Prolog).

```
penguin(fido).
penguin(fido).
penguin(fido).
... (repeated to 512 tokens)
query(penguin(fido)).
```

ISR interaction. ISR ≈ 0. Each copy is individually redundant since identical copies remain after removal. ISR correctly diagnoses this degeneration.

Prevalence. Observed in FOL exec-only seed 43 (87% of outputs hit max completion length; PR = 13%, ISR = 0).

D.7 Complexity-without-Utility

Definition. The formalization contains syntactically complex premises (nested quantifiers, multi-arity predicates) that are semantically vacuous: none individually necessary for the proof. The complexity inflates surface metrics (formula length, nesting depth) without contributing to logical derivation.

Example (FOL).

```
Premises:
  all x all y (R(x,y) ->
    exists z (S(x,z) & T(z,y))).
  all x (P(x) -> Q(x)).
  P(a).            % only this needed
Conclusion: Q(a).
```

ISR interaction. ISR correctly identifies the vacuous premises (ISR = 0 for each), pulling down the mean ISR. This is the mode ISR is best calibrated for.

Prevalence. Low. Observed in Random β=0.1 seed 42 at 3.8% (3/79). Not observed under ISR reward (ISR's gradient discourages zero-ISR premises).

D.8 Contradiction Explosion

Definition. The component count grows unboundedly over training. Excess premises introduce contradictions or redundancies that trivially entail any conclusion via *ex falso quodlibet*. Individual premises may register as ISR = 0 (removable without breaking the proof), but the sheer count obscures the underlying triviality.

Example (FOL, 1000-step exec-only).

```
Premises:
  c1. c2. c3. ... c18. % 18.55 mean
  % conclusion embedded among premises
Conclusion: Q(a).
```

ISR interaction. ISR ≈ 0 (at 1000 steps: ISR = 0.003, with 94.8% of proved samples having ISR = 0).

Prevalence. Observed in extended exec-only training (seed 42, 1000 steps). Mean component count grows from $\sim$6 (500 steps) to 18.55 (1000 steps).

D.9 Query-as-Fact Gaming

Definition. The model converts the natural language conclusion into a snake_case atom and includes it as both a ground fact and the query. Prolog trivially proves the query by unifying with the fact. This mode evades text-match classifiers designed for CE/TS because the generated atom bears no surface resemblance to the original conclusion text.

Example (Prolog).

```
% NL: "Fido is happy."
% Model output:
fido_is_happy.
:- fido_is_happy.      % query = fact
```

ISR interaction. ISR ≈ 0.9–1.0. The single fact is trivially necessary (its removal breaks the query), inflating ISR. Functionally equivalent to TS but with a distinct surface mechanism.

Prevalence. Observed in Prolog exec+STA seed 44 (PR = 0.97, ISR = 0.912). 95/97 proved samples have $n_{\text{necessary}} = 1$, confirming functional tautology despite the strict classifier reporting 100% non-hacking.

E Per-Seed Detailed Results

Tables 7 and 8 report full per-seed results. All metrics are computed by `eval_sta.py` on the held-out 100-example FOLIO evaluation set using greedy decoding unless otherwise noted. ISR values are proved-only ISR*.

ISR collapse trajectory (FOL exec-only, seed 42). Table 7 traces the ISR collapse over training: ISR* = 0.642 at 100 steps → 0.132 at 500 steps (−79.4%) → 0.003 at 1000 steps (−99.5% from step 100). PR increases monotonically (0.32 → 0.93 → 0.96) while component necessity collapses, confirming that ISR flags degeneration invisible to execution reward. By step 1000, 94.8% of proved samples have ISR = 0 and the mean component count rises to 18.55.

Prolog: universal Goodhart via three distinct pathways. All three Prolog exec+ISR seeds exhibit Goodhart's Law with 0% non-hacking proofs (Figure 2), but each discovers a different gaming mechanism: seed 42 produces clean TS tautologies (single premise = conclusion), seed 43 generates uppercase pseudo-Prolog that bypasses the text-match classifier while remaining functionally tautological, and seed 44 converts NL conclusions to snake_case atoms used as both fact and query. The diversity of gaming pathways across seeds (despite identical training data and reward function) demonstrates that Goodhart's Law in neurosymbolic reasoning is not a single failure mode but a family of convergent exploits.

E.1 Per-Seed Results at β=0.2 and β=0.3

Table 10 reports per-seed results for ISR at β=0.2 and both ISR and Random at β=0.3. At β=0.2, all three seeds converge to pure TS Goodhart (100% or near-100% TS), confirming $\beta \geq 0.2$ as the extremal Goodhart regime. The three ISR β=0.3 seeds illustrate qualitatively distinct Goodhart pathways (§4.3).

E.2 Early-Training Hacking Analysis (100-Step Checkpoint)

To control for the prove-rate confound in our SFT comparison (§4.2), we analyze the exec-only checkpoint at 100 training steps (seed 42), where PR=32%, higher than SFT (13.0%, 3-seed pooled) but still early in training. Of the 32 proved samples, 9 (28.1%) exhibit TS hacking (single premise identical to conclusion) and 1 (3.1%) exhibits CE (conclusion embedded among premises), yielding a total hacking rate of 31.2%. By contrast, SFT at comparable or lower PR shows 7.7% hacking (1/13 proved samples). This demonstrates that reward hacking emerges early in GRPO training, well before the prove-rate plateau, and is not simply an artifact of having more proved samples available for exploitation. The 100-step hacking rate (31.2%) is already substantially above SFT (7.7%), supporting the claim that hacking is associated with RL optimization rather than prove-rate level alone.

E.3 $O(1)$ vs. $O(n)$ ISR Ablation

All main experiments (§4.2–4.4) use $O(n)$ full LOO ISR during both training and evaluation; the $O(1)$ binary variant described below exists as an alternative implementation (`execution_fast_sta`) but is used only in the ablation reported here. The two signals measure fundamentally different properties, not approximations of the same quantity. The binary signal tests *collective* premise necessity: fast_ISR(x) = 1 if x is proved and the conclusion is not provable under empty premises (tautology check), 0 otherwise. LOO tests *individual* premise necessity: each component is

removed in turn, and ISR reflects the fraction whose removal causes proof failure.

Evaluation-time comparison. Across 402 proved samples from all ProofFOL conditions, the $O(1)$ binary signal is 1.0 for every sample: non-tautological conclusions cannot be proved without any premises, so the collective necessity test is trivially satisfied. This renders $O(1)$ unable to distinguish hacking from genuine formalizations: both pass the empty-premise check. By contrast, $O(n)$ LOO produces a clear separation: exec-only models average ISR*=0.046 (n=234) while ISR-trained models average ISR*=0.183 (n=168), because hacking constructions contain individually redundant components that LOO detects. Of 179 disagreements between the two signals, all are false positives ($O(1)$=1, LOO=0): the binary check approves formalizations that LOO correctly identifies as structurally degenerate.

Training-time ablation. As a training reward ($r = 0.5\, r_{\text{exec}} + 0.5\, \text{fast_ISR}$, seed 42, 500 steps), the $O(1)$ variant achieves PR=63% with ISR*=0.007, functionally zero, compared to ISR*=0.132 under exec-only and ISR*=0.42 under $O(n)$ ISR β=0.1. The tautology check depresses prove rate by 30 pp relative to exec-only (63% vs. 93%) by penalizing formalizations where the conclusion is provable without premises, but this constraint provides no gradient toward individual component necessity; the model is free to degenerate as long as it avoids trivial tautologies. That even the fine-grained $O(n)$ LOO signal, which *does* provide per-component necessity feedback, is ultimately gamed (§4.3) underscores that the problem lies in reward optimization itself, not in signal granularity.

Computational overhead. Full LOO requires n+1 solver calls per sample (mean $n\approx6$ for FOLIO). With Prover9 at ∼5 ms per call, STA evaluation adds ∼55 ms per sample versus ∼4 s for LLM inference (<1% overhead). The $O(1)$ binary check adds a single solver call per proved sample (<2% per training step), but the evaluation above shows this saving comes at the cost of all diagnostic power.

E.4 Cross-Model Generalization: LLaMA-3.2-3B Cold-Start Failure

We applied the identical GRPO configuration (Table 3) to LLaMA-3.2-3B (Grattafiori et al., 2024) (3B; exec-only, seed 42, K=4, FOLIO). Across 72 of 500 planned steps the model produced zero valid Prover9 output, generating only natural-language text; r_{exec}=0 for all rollouts yields zero GRPO advantage, so training was terminated early.[5] This establishes a *capability threshold*: the base model must produce syntactically valid solver input for the RL loop to generate signal. Qwen3-4B crosses this threshold (non-zero prove rates within 10 steps) while LLaMA-3.2-3B does not despite comparable size (3B vs. 4B), likely reflecting pretraining data composition. An SFT warmstart (3 epochs on gold formalizations, lr=2×10^{-5}) achieves PR=16% (n=100) after correcting the generation-termination issue. However, the model produces Unicode FOL notation (∀, →, ∧) instead of the required ASCII syntax despite training on ASCII-formatted data; the 16% prove rate reflects Prover9's partial Unicode tolerance rather than correct format adherence. Subsequent GRPO runs (exec-only, 500 steps from SFT warmstart, 3 seeds) fail to improve beyond SFT baseline (PR=15.0±1.0%, n=14–16 proved per seed), yet produce no structural degeneration: 86.6±0.9% of proved samples are genuine across seeds, with cross-seed variance near zero (σ_{PR}=1 pp). This contrasts sharply with Qwen3-4B, where GRPO reaches 93% PR but collapses to 8% genuine with high cross-seed variance. The result suggests structural degeneration is a byproduct of successful RL optimization, not of the training procedure *per se*: when GRPO fails to amplify prove rate, the model retains structural integrity.

E.5 Seed-Dependent Convergence in Exec-Only GRPO

Not all exec-only seeds converge. Of 6 seeds tested (s42–s46, s52), we observe three regimes: *converging with degeneration* (s42: PR=93%, genuine=7.5%; s44: PR=96%, genuine=0%; s46: PR=93% at

[5] This run was also affected by a generation-termination configuration issue (dual EOS tokens); the zero prove rate reflects both the base model's lack of FOL generation capability and truncated outputs.

500 steps), *slow convergence with degeneration* (s43: PR=14% but 100% hacking at 500 steps), and *non-converging* (s45: PR=15%, genuine=86.7%; s52: PR=14%, genuine=85.7%). Non-converging seeds retain high genuine rates (85–87%), indicating that structural degeneration is conditional on successful RL optimization: when GRPO fails to bootstrap prove rate beyond base-model levels, the model preserves structural integrity. Extended training to 1000 steps for s45 confirms persistent non-convergence (PR=15%, ISR*=0.592, genuine=86.7%, n=15 proved); the model neither improves nor degenerates.

This parallels the LLaMA-3.2-3B finding (Appendix E.4), where GRPO also fails to improve PR and retains structural quality. Both cases support the interpretation that degeneration is a byproduct of reward optimization success, not of the training procedure per se.

E.6 Cross-Scale: Qwen3-8B Exec-Only GRPO

To test whether structural degeneration generalizes beyond 4B scale, we apply exec-only GRPO to Qwen3-8B with K=8 rollouts per prompt (K=4 produces near-identical completions at 8B scale, yielding zero GRPO advantage and no learning signal). Table 11 summarizes configuration differences.

We first tested K=4 (matching the 4B configuration): the 8B model produces near-identical rollouts across all four completions, yielding zero GRPO advantage throughout 500 steps and no learning signal (PR=15.5%, cold-start stagnation). Increasing to K=8 provides non-zero reward variance during mid-training (approximately 40–50% of steps between 20–370 show non-zero advantage), but this signal proves insufficient for convergence: held-out evaluation yields PR=14% (n=14 proved), ISR*=0.573, genuine=92.9% (13/14), with a single CE sample. Cosine learning-rate decay combined with insufficient early bootstrap appears to prevent the model from escaping the low-PR regime. Table 12 summarizes the cross-scale comparison.

This result mirrors two other non-convergence cases: (1) the seed-dependent 4B non-convergence (s45, s52; Appendix E.5), where identical 4B configurations fail to bootstrap, and (2) the LLaMA-3.2-3B cold-start failure (Appendix E.4), where the base model lacks FOL generation capability entirely. Together, these cases establish a convergence threshold for GRPO-driven degeneration: the structural collapse documented in §4.2 requires successful RL optimization as a prerequisite, and at 8B scale the rollout diversity bottleneck prevents this optimization from occurring under our configuration.

Responsible NLP Research Checklist

A. For every submission:

A1. **Did you describe the limitations of your work?** Yes. Section 6 discusses single model family and dataset, ISR's syntactic-only nature, limited seed counts, classifier coverage gaps, data scale, and temporal-snapshot caveats.

A2. **Did you discuss any potential risks of your work?** Yes. Section 7 discusses implications for AI safety, specifically that metrics designed to detect misalignment may themselves become optimization targets.

A3. **Do the abstract and introduction summarize the paper's main claims?** Yes. Both summarize the two contributions (C1–C2) and key quantitative findings.

A4. **Have you used AI writing assistants when working on this paper?** Yes. AI assistants were used for drafting and editing. All scientific claims, experimental results, and analyses were verified by the authors.

B. Did you use or create scientific artifacts? Yes.

B1. **Did you cite the creators of artifacts you used?** Yes. We cite FOLIO (Han et al., 2024), Qwen3 (Qwen Team, 2025), GRPO (Shao et al., 2024), Prover9 (McCune, 2005–2010), and SWI-Prolog (Wielemaker et al., 2012).

B2. **Did you discuss the license or terms for use and/or distribution of any artifacts?** Yes. Qwen3 is released under

the Tongyi Qianwen License; FOLIO is released under CC BY-SA 4.0. Both permit research use.

B3. **Did you discuss if your use of existing artifact(s) was consistent with their intended use?** Yes. FOLIO is used for its intended purpose (NLI evaluation); Qwen3 is used within its intended scope (language model fine-tuning).

B4. **Did you discuss the steps taken to check whether the data that you are using/creating contains any information that names or uniquely identifies individual people or offensive content?** Not applicable. FOLIO contains logical reasoning problems without personal information or offensive content.

C. Did you run computational experiments? Yes.

C1. **Did you report the number of parameters in the models used, the total computational budget, and computing infrastructure used?** Yes. The model size (4B) is reported in §3.1. Training details (GRPO K=4, 500/1000 steps, seeds, batch sizes) are in §3.1 and Appendix A. Computing infrastructure and total GPU hours are reported in Appendix A.

C2. **Did you discuss the experimental setup, including hyperparameter search and best model selection strategies?** Yes. All hyperparameters are reported in §3.3 and Appendix A. No hyperparameter search was performed; β values were chosen for dose-response analysis.

C3. **Did you report descriptive statistics about your results (e.g., error bars, confidence intervals)?** Yes. We report mean±std across seeds (Table 1), per-seed ranges, bootstrap confidence intervals (Appendix Table 5), and discuss statistical power limitations (§6).

C4. **If you used existing packages (e.g., for pre-processing, models, or evaluation), did you report the implementation, model, and version numbers?** Yes. We report Qwen3-4B model versions, Prover9, and SWI-Prolog. Training uses TRL 0.17.0, PyTorch 2.6.0, and Transformers 4.51.3; versions are listed in Appendix A.

D. Did you use human annotators (e.g., crowdsourcing or research with human participants)? Minimal. Manual audit was conducted for classifier precision (50 random CE/TS samples verified; Appendix D) and for PG false-negative correction (seed 42 only). No crowdsourced annotation was used.

E. Did you use AI assistants (e.g., ChatGPT, Copilot, or Code Interpreter)? Yes. AI assistants were used for code development and manuscript preparation. All experimental designs, analyses, and scientific conclusions were verified by the authors.

Table 7: Detailed per-seed results: FOL (Prover9). Includes the per-seed β=0.1 breakdown referenced from Table 1. PR = prove rate, ISR* = proved-only ISR, TS/CE/N-H = percentage among proved samples. "—" indicates data not available. †s43 uses sampling eval (T=0.7). ‡Three s44-1000 samples escape text-match heuristics (CE/PG classifiers) but exhibit ISR=0 and functionally degenerate proof structures: disjunctive conclusions that bypass PG detection and commuted premises that bypass CE detection; reclassified as degenerate. ◇Seeds 45–46 added post-hoc under identical hyperparameters (effective batch size 8; micro-batch decomposition differs: 4×2 vs. 2×4, mathematically equivalent).

Condition	**Seed**	**Steps**	**PR**	**ISR***	**TS%**	**CE%**	**N-H%**	**Notes**
Baselines								
Base model (no training)	—	0	0.12	0.064	—	—	—	Pre-training
Ground-truth FOL	—	—	0.18	0.559	—	—	—	Human formalizations
SFT	42	1000	0.13	0.568	0	7.7	**92.3**	Supervised, no RL
	43	1000	0.11	0.635	0	9.1	**90.9**	
	44	1000	0.15	0.589	0	6.7	**93.3**	
	3-seed pooled		0.130	0.595	0	7.8	**92.3**	n=39 proved
Exec-only								
	42	100	0.32	**0.642**	—	—	—	Early checkpoint
	42	500	0.93	0.132	0	87.1	7.5	PG=6.5%
	42	1000	0.96	0.003	0	100	0	Contradiction explosion
	43	500	0.13	0†	—	—	—	†Greedy (n_{proved}=13)
	43	1000	0.96	0.082	0	93.8	6.2	CE dominant
	44	500	0.98	0.582	100	0	0	Tautological collapse
	44	1000	1.00	0.002	0	85	0‡	PG=15%
	3-seed mean±std		0.68±0.47	0.238±0.30				500-step
	3-seed mean±std		0.97±0.02	0.029±0.05			2.1±3.6	1000-step
ISR β=0.1								
	42	500	0.81	0.415	16.0	51.9	**17.3**	Mixed CE+TS+PG
	42	1000	0.90	0.877	96.7	0	2.2	TS collapse; PG=1 (n=90)
	43	500	0.61	0.830	**100**	0	0	TSa=59, TSb=2
	44	500	0.89	0.902	100	0	0	TS dominant (FN corrected)
	45◇	500	0.42	0.605	38.1	4.8	**54.8**	PG=1; n=42
	46◇	500	0.77	0.411	14.3	26.0	**54.5**	PG=4; n=77
	3-seed mean±std		0.77±0.14	0.716±0.26				500-step (s42–44)
	5-seed mean±std		0.70±0.19	0.633±0.23				500-step (s42–46)
ISR β=0.3								
	See Table 10 for per-seed β=0.3 results.							
Random β=0.1								
	42	500	0.79	0.131	0	73.4	22.8	PG=4
	43	500	0.94	0.150	**38.3**	59.6	1.1	TSa=36; PG=1
	44	500	0.89	0.070	0	82.0	6.7	PG=16 (CE∩PG: 6)
	3-seed mean±std		0.87±0.08	0.117±0.042				TS seed-dep.: 0/38/0%
Random β=0.3								
	42	500	0.48	0.094	8	92	0	TSa=4 (D086); n_{comp} = 9.6
	43	500	0.92	0.180	7.6	89.1	3.3	TSa=7; $n_{\text{N-H}}$=3
	44	500	0.23	0	0	100	0	Pure CE (n=23)
	3-seed mean±std		0.54±0.35	0.09±0.09				
Shuffled-ISR β=0.1 *(deconfounding control)*								
	42	500	0.15	0.567	6.7	6.7	**86.7**	n=15 proved
	43	500	0.16	0.568	6.3	6.3	**87.5**	n=16 proved
	44	500	0.15	0.542	6.7	6.7	**86.7**	n=15 proved
	3-seed mean±std		0.153±0.006	0.559±0.015	6.5	6.5	**87.0**	N=46 proved

Table 8: Detailed per-seed results: Prolog (SWI-Prolog 8.4.2). PR = prove rate, ISR* = proved-only ISR, TS/CE/N-H = percentage among proved samples. "—" indicates data not available.

Condition	Seed	PR	ISR*	TS%	CE%	N-H%	Notes
Exec-only							
	42	0.83	0.037	0	0	—	Ground fact enumeration
	43	0.56	0.015	—	—	—	
	44	0.21	0.387	—	—	—	Small-sample (n=21)
	3-seed mean±std		0.53±0.31	0.146±0.21			
Exec+ISR (STA reward)							
	42	0.50	1.000	**100**	0	0	TS tautological
	43	0.38	0.185	0‡	97.4	0	‡Functional TS[a]
	44	0.97	0.912	0‡	0	0‡	Query-as-fact gaming[b]
	3-seed mean±std		0.62±0.31	0.699±0.45			3/3 Goodhart, 0% non-hacking

[a] Strict TS classifier reports 0%, but functional analysis shows all 38/38 proved samples are tautological ($n_{\text{necessary}} = 1$). Uppercase pseudo-Prolog syntax bypasses the text-match classifier.
[b] Strict classifier reports Non-hacking = 100%, but 95/97 proved samples have $n_{\text{necessary}} = 1$. The model converts NL to snake_case atoms serving as both fact and query, a third distinct Goodhart pathway.
‡ Strict classifier value is misleading; see footnotes [a]/[b].

Table 9: Detailed per-seed results: ProofFOL (Prover9). PR_{f50} = mean prove rate over final 50 steps. TS = tautological shortcutting (§4.5), CE = conclusion embedding, Genuine = classified as neither TS nor CE. Stable convergence: PR_{f50}>0.90.

Condition	Seed	PR_{f50}	ISR*	TS%	CE%	Genuine%	Notes
Base	—	0.16	0.310	0	0	**100**	n=16 proved
Exec-only							
	42	≈1.0	0.066	**100**	0	0	n=99; stable
	43	0.55	0.119	0	49.2	50.8	n=63; partial
	44	≈1.0	0.043	0	98.6	1.4	n=72; stable
	Aggregate			17.4% genuine (mean of seeds)			
ISR β=0.1							
	42	0.35	0.12	0	0	**100**	n=46; collapsed
	43	≈0.47	≈0.13	0	69.6	30.4	n=56; unstable
	44	0.49	0.145	0	0	**100**	n=66; oscillation
	Aggregate			**76.8%** genuine (mean of seeds)			

Aggregate genuine rate: mean of per-seed genuine rates. Permutation test (exec vs. ISR genuine): p=0.10, N=3 per group.

Table 10: Per-seed results at β=0.2 and β=0.3 (FOL). PR = prove rate, ISR* = proved-only ISR, TS/CE/N-H = percentage among proved samples.

Condition	Seed	PR	ISR*	TS%	CE%	N-H%	Mode
ISR β=0.2							
	42	1.00	0.602	**100**	0	0	Pure TS (n=100)
	43	0.96	0.552	**96.9**	1.0	0	PG=2
	44	0.94	0.964	**100**	0	0	Pure TS (n=94)
	3-seed mean±std		0.97±0.03	0.71±0.22			3/3 TS Goodhart
ISR β=0.3							
	42	0.18	0.143	0	77.8	5.6	Reward collapse (n=18); PG=3
	43	0.86	0.295	4.7	68.6	23.3	CE shortcutting
	44	0.95	0.742	**100**	0	0	Pure TS Goodhart
	3-seed mean±std		0.66±0.42	0.39±0.31			
Random β=0.3							
	42	0.48	0.094	8	92	0	TSa=4 (D086)
	43	0.92	0.180	7.6	89.1	3.3	TSa=7; $n_{\text{N-H}}$=3
	44	0.23	0	0	100	0	Pure CE (n=23)
	3-seed mean±std		0.54±0.35	0.09±0.09			

	Qwen3-4B	Qwen3-8B
Rollouts (K)	4	8
Batch size	4	1
Gradient accum.	2	16
Effective batch	8	16

Table 11: Configuration differences between 4B and 8B GRPO runs. K=8 is required at 8B scale because K=4 produces near-identical rollouts, eliminating gradient signal.

Config	PR%	Genuine%	Converged?	Gradient
4B K=4 s42	93	7.5	Yes (degenerate)	Sufficient
8B K=4 s42	15.5	—	No	Zero
8B K=8 s42	14	92.9	No	Partial

Table 12: Cross-scale comparison. The 4B model converges with structural degeneration; both 8B configurations fail to converge, preserving structural integrity. K=8 provides partial gradient signal absent in K=4 but insufficient for convergence within 500 steps.